Towards Demographic-Based Photographic Aesthetics Prediction for Portraitures

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Abstract. Do women look at aesthetics differently from men? Does cultural background have an influence over the perception of beauty? Has age have any role in this? Psychological and art studies reveal striking differences in perception among various demographical aspects. This warrants attention particularly with the rapid growth in automatic evaluation of photo aesthetics. In this research, we investigate the influences of demographic factors of photographer towards the aesthetic quality of portrait photos from the computational perspective. An extended version of the large-scale AVA dataset was created with the inclusion of the photographers' demographic data such as location, age and gender. We trained several demographic-centric CNN models, which are then fused together as a single multi-demographic CNN model to learn aesthetic tendencies in a holistic manner. We demonstrate the efficacy of our model in achieving state-of-the-art performance in predicting portraiture aesthetics.

1 Introduction

In a recent study on beauty and the brain, Jacobsen [1] revealed that influences of demographic attributes such as history, culture, and individual differences are important determinants of aesthetics appreciation. In a particular eye-opening work, Masuda et al. [2] studied the aesthetics preferences of East Asians and Americans for portrait photographs and concluded that the Western societies prefer object-oriented attention whereas Ease Asians desire context-oriented attention in portraiture. Apart from cultural differences, Jacobsen [1] has highlighted that within cultures, individual differences such as age, gender, expertise (training in art), and education also influence aesthetics perception. Taunton [3] investigated the influence of age on preferences for subject matter, realism, and spatial depth in painting reproductions and found significant difference in preferences across different age group. In another interesting study, Cela-Conde et al. [4] infer that men and women use different strategies in assessing aesthetic preferences, where men focus on big picture and women pay more attention to local details. All these studies support Jacobsen's [1] findings that visual aesthetics perception is highly complex and is supported by highly-configured dynamic neural networks.

With the explosive growth of digital mobile cameras and social media that leads to vast amount of photographs being captured and shared everyday, computational aesthetics is receiving increasing research attention. Many state-of-the-arts aesthetics prediction models focus on classifying photographs as either aesthetically high or aesthetically low. Earlier traditional approaches utilize handcrafted features [5–8] or generic descriptors [9] to predict aesthetic quality. More recently, with the availability of the large-scale Aesthetics Visual Analysis (AVA) dataset [10], researchers explored deep learning approaches [11,12] which were successfully in outperforming traditional methods by a good measure. Notably, most of these works focus on predicting aesthetics of generic photos; very few research works investigated the aesthetics of a specific category of photos such as portraits [13–15]. In addition, despite the significant influence of demographic attributes on aesthetics, little research [14] has considered capitalizing on these attributes to improve aesthetics prediction.

In this paper, we aim to investigate the influence of photographers' demographic profiles on aesthetics prediction of portrait images, from the computational perspective. As photographers with different demographic profiles have the tendency to perceive aesthetics differently, they may likely use different photographic techniques to capture and enhance their photographs. By identifying three demographic aspects of photographers: location, age, and gender, we first categorized the photos from AVA-Portraits dataset [14] based on newly extracted demographic information. Location is a notion closely related to culture and it can thus be an indicator of the influence of culture on aesthetics. We build a pair of convolutional neural network (CNN) models for each demographic aspect to capture intrinsic tendencies. We then combine these individual demographic-based CNNs (dgCNN) into a multi-demographic-based CNN (MdgCNN) as illustrated in Fig. 1. Experimental results demonstrate that our

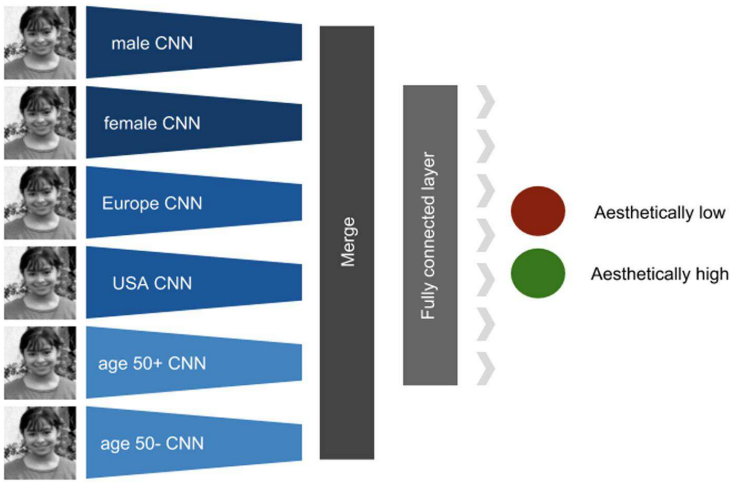

Fig. 1. Combining multiple demographic-centric CNNs for aesthetic prediction of portraitures.

proposed idea of learning demographic-centric models is one that is worth considering for subjective tasks such as the perception of aesthetics.

2 Related Work

Majority research in computational aesthetic focuses on investigation of generic photos. Little research has been performed on portrait photos. One of the earlier works that focus on portrait aesthetics [13] uses a feature set that combines template-based spatial composition features and highlight/shadow composition. According to the authors, the spatial composition features are more discriminative than highlight/shadow composition features.

Redi et al. [14] extracted a set of hand-crafted visual features based on compositional rules, semantics and scene content, basic quality metrics, portrait specific features and fuzzy properties. They highlighted that the main features that correlates with photo aesthetics are sharpness of facial landmark, contrast, exposure, uniqueness and originality. Their proposed model outperformed the method of Khan and Vogel [13] by 14.66% on the same dataset. Interestingly, their results show that demographic factors like race, gender and age of the person who are in the photo are uncorrelated with photo aesthetics. We believe that likely the correlation of culture, gender and age with photographic aesthetics lies on the photographers, and not in the person captured in the photo.

More recently, Kairanbay et al. [16] proposed a new model that employed the GoogleNet CNN model [17] with data augmentation. In their approach, the crops from the four corners and the face are feed forwarded to the GoogleNet and the layer before each softmax layer is used as the feature vector. These five feature vectors from the five crops is then concatenated to form the final feature vector and passed to a linear classifier for learning. The usage of the five crops helped to mimic the foreground-background compositional rules, which possibly contributed to the improvement in classification accuracy. However, the accuracy of their deep learning solution is only marginally better than Redi et al.'s [14] hand-crafted solution.

3 Dataset

The AVA dataset [10] is the benchmark dataset which is widely used in the photo aesthetic task. The dataset consists of 250K photos, where 230K photos are used for training and the remaining 20K photos are used for testing. Each photo is evaluated by an average of 210 users with an aesthetic score that ranges from 1 to 10. AVA dataset also provides other meta-data such as semantic tags, challenge ID and style information for each photo.

Redi et al. [14] extracted a subset of portrait images from the AVA dataset by filtering out photos that contain "Portrait" semantic tag and challenges that contain either one of these words: "Portraiture", "Portrait" and "Portraits" As some of the extracted photos do not contain real portraits, Face++ API [18] is used to filter out the non-portrait images. The final dataset (also coined as "AVA-Portraits" in [16]) consists of a total of 10,141 portrait photos.

3.1 AVA-Portraits+

The original AVA dataset does not have any information of the photographers' demographic profile that is crucial in this research. Therefore, we set out to extend the AVA-Portraits data by including additional data of three demographic aspects: location, age and gender. We name this extended dataset, the "AVA-Portraits+". The demographic profile for each photographer is extracted from the DPChallenge web site, the original source of the AVA.

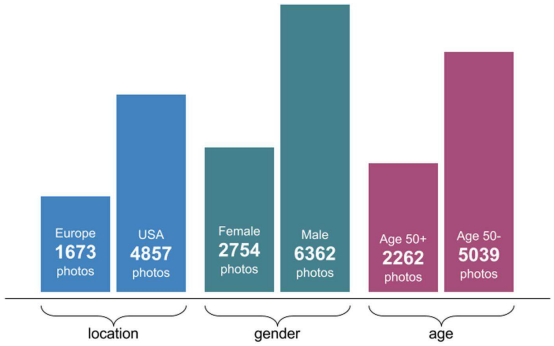

Fig. 2. The number of photos for the image subset of each demographic attribute in AVA-Portraits+.

Figures 3, 4a and b illustrate the distribution of photographer locations, age, and gender respectively on the full AVA dataset. From Fig. 3, we can observe that USA (North America) and Europe are the two continents with significantly larger number of photographers in DPChallenge. Due to lack of data from other continents, our study on location is limited only to USA and Europe. We create a bounding box for each continent and selected only photos from inside these bounding boxes (See Fig. 3, hence omitting photos from other continents.

For the age data, Fig. 4a shows that there are some photographers whose age are more than 100 or less than 5. In our experiments we ignore these outliers and consider 15 and 85 as the minimum and maximum ages respectively. The threshold is computed as the average of the minimum and maximum values, i.e. 50. We divide the dataset into two subsets: the first consists of images taken by the photographers whose age is more or equal than 50, and the second subset contains images captured by photographers whose age is less than 50.

Meanwhile, Fig. 4b shows that the number of male photographers are 2.3 times more than female photographers. Whilst the overall average aesthetic score for males (5.45) is slightly higher than that for females (5.36), they are very close to the mean aesthetic score (5.5) of the AVA-Portraits dataset. Figure 2 summarizes the amount of photos for each demographical subset of the AVA Portraits+ dataset. Similar to that of the location and age data, the dataset is also divided into two subsets for the male and female photographers.

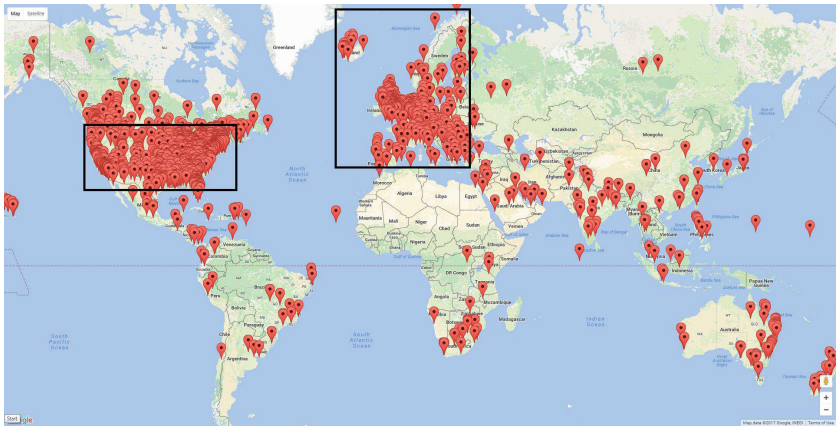

Fig. 3. The distribution of photographers' location in AVA dataset. Bounding boxes indicate the two regions where photos were selected.

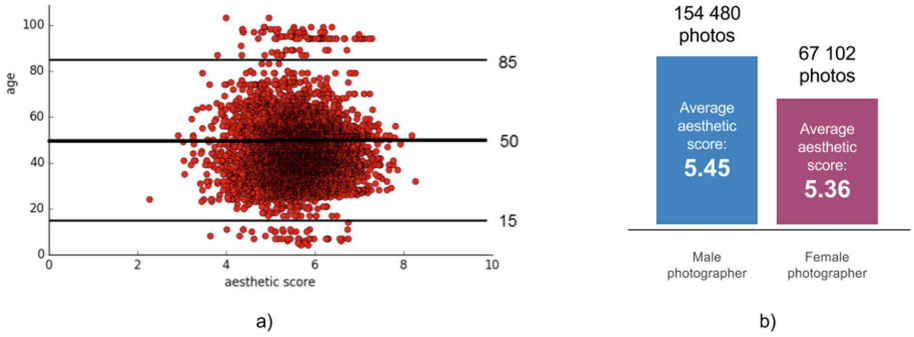

Fig. 4. (a) The distribution of photographers' age in AVA dataset. 85 and 15 are the upper and lower bounds, (b) The distribution of photographers' gender in AVA dataset.

Photographer-subject correlations. For further investigation into the AVA portraits+ dataset, we explore the correlation between the photographer and the subject in the photo (whose portrait was taken). Using the Face++ API [18] service, we detected the face region and demographic parameters (e.g. gender, race and age) of the subject in the photo. Table 1 depicts the demographic matches between the photographer and the subject in the photo; the numbers denote the percentage of portraits with the column attribute captured by a photographer with the row attribute. The rows indicate the photographer's demographic profile while the columns represent the subject's demographic profile. Analysis of Table 1 reveals that females often take photos of females than males, while European photographers tend to shoot portraits that are not necessarily smiling.

Table 1. Demographic attribute matching between photographer and subject in the AVA-Portraits+ photos.

	Male (%)	Female (%)	White (%)	Asian (%)	Black (%)	Smiling (%)	Avg. age
Europe	44.76	55.23	76.53	18.3	5.15	32.07	38.66
USA	43.2	56.79	76.25	17.86	5.87	39.07	36.51
Male	46.04	53.95	73.75	19.38	6.86	35.55	37.67
Female	37.02	62.97	77.11	17.69	5.18	39.42	35.53
age 50+	43.67	56.32	74.65	18.97	6.37	39.6	39.01
age 50−	42.45	57.54	75.86	18.86	5.27	35.97	35.42

4 Learning Demographic-Centric Models

AlexNet [19] is a well known deep Convolutional Neural Network (CNN) model that is effective at solving many complex computer vision tasks. Unlike its 'deeper' successors such as GoogleNet [17], the AlexNet consists of five convolutional layers and two fully connected layers followed by a final softmax layer. In this work, we opted for AlexNet as our base model as the data in this problem is only around 10K images and could very easily overfit a much deeper model. We hypothesize that finer grained features can be learned from specific subsets of a dataset, which can lead to better representation for subjective tasks such as photo aesthetics. We intend to uncover further understanding into whether better models can be learned if the photographers' intrinsic background properties can be untangled. As such, we propose to train demographic-centric CNN models on the subsets of the AVA-Portraits+ dataset to capture the natural inclinations of photographers based on their demographics profile.

4.1 Baseline Methods

The two handcrafted methods [13,14] which were reported in [14] are used as the vanilla baselines for AVA-Portraits. We also establish a couple of deep learning baselines, which are based on transfer learning.

Firstly, we fine-tuned the ImageNet trained AlexNet CNN model [19], by replacing the final softmax layer with one that is of output dimension 2×1, corresponding to the two aesthetic classes (high and low). We use $\delta = 1.5$ setting[1], which results in around 16K photos selected for training. A classification accuracy of 75.15% was obtained on the test partition (20K photos) of the original AVA test set [10]. On the AVA-Portraits dataset, this fine-tuned CNN model attained a classification accuracy of 63.8% on a threshold value of 5.5 based on its mean rating [14], instead of 5.0 used in AVA. For fair comparison, we followed

[1] The authors of the AVA dataset [10] introduced the parameter δ to filter ambiguous photos during the training step; photos with average rating $[5-\delta, 5+\delta]$ are omitted. $\delta = 0$ in the testing step.

the same evaluation protocol and train-test partitions provided by the authors of the AVA-Portraits dataset [14].

In our second baseline, we took the first model and further fine-tuned it with only portrait photos (from AVA-Portraits) and yielded a 68.61% accuracy, which outperformed the baseline solutions [13,14] by around 4%, and also a recent work by [16] which employed deeply learned features but without transfer learning. Table 2 shows the classification accuracy of the baseline approaches and their corresponding dispersion errors. At this juncture, results showed that transfer learning is an appropriate technique to learn the aesthetics of portraits better.

Table 2. Aesthetic classification accuracy and dispersion error of the proposed methods compared against baseline methods.

Type	Methods	Accuracy (%)	Error ± (%)
HC	Khan et al. [13]	54.58	1.25
HC	Redi et al. [14]	64.24	1.76
DL	Kairanbay et al. [16]	65.33	1.01
DL	Fine-tuned CNN (AlexNet)	63.80	1.33
DL	Doubly-Fine-tuned CNN (AlexNet)	68.46	1.95

*HC: Handcrafted methods; DL: Deep learning methods

4.2 Demographic CNNs

The new AVA-Portraits+ dataset enables us to learn *demographic-based CNN* models, or **dgCNN** (in short from now on). For the six demographic attributes identified (see Fig. 2), we train a single dgCNN for each attribute individually, using samples from their respective subsets. In similar fashion to the transfer learning procedure reported earlier, we fine-tune our best model ("Doubly-Fine-tuned CNN") from Table 2 with samples from each attribute subset. A 5-fold cross validation approach is used for the evaluation of performance of the dgCNNs. Figure 6 reports the classification accuracy of each dgCNN and its corresponding dispersion error.

To ascertain the benefits of our demographic-based sampling strategy, we randomly extracted the same number of photos from AVA, captured by photographers of the particular demographic for training purpose. We made sure there were no overlaps with the test portrait photos selected for each fold. This time round, the individual dgCNN models are trained with the randomly selected photos on the same network parameters and test set (per fold).

4.3 Multi-demographic CNNs

The demographic attributes learned from the dgCNNs can be fused together into a multi-demographic CNN model, or **MdgCNN** (in short from now on). As a photographer's profile can be of multiple demographic attributes (gender,

age, location), we intend to capture the complex individual inclinations towards the aesthetic quality of portraits, all through one integrated two-level network.

Motivated by brain-inspired deep networks from [20], we attempt to learn the demographic attributes along a selected feature dimension independently by treating these dgCNNs as "parallel pathways", a concept derived from visual neuroscience. Using the individually pre-trained dgCNN models, we freeze the weights from all convolutional layers (1st to 5th) and only the fully connected (FC) layers are backpropagated during training. The activations at the respective FC layers (6th or 7th) of each dgCNN are taken as the learned demographic attributes. We merge (by concatenating) these activations together, before connecting to another FC layer which provides high-level association between features picked up at the dgCNN level and aesthetic quality. Figure 5 illustrates the architecture of the proposed model. We further experimented with an additional drop-out layer after the FC layer that is selected for merging, in order to introduce some regularization during training. All images have been feed forwadred in testing stage without taking into account its demographic attributes.

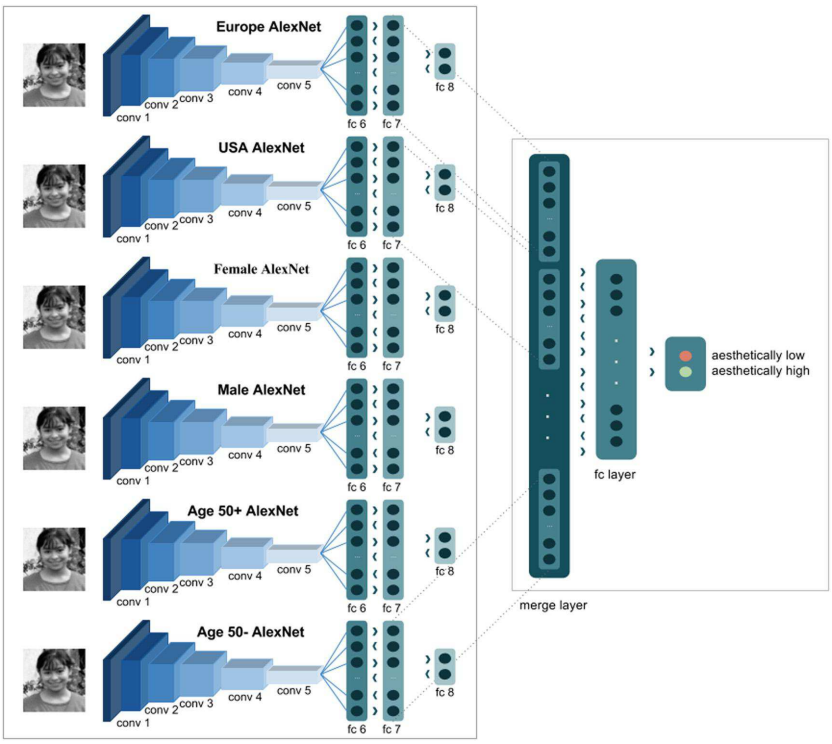

Fig. 5. The architecture of the proposed MdgCNN model

5 Results and Discussion

Figure 6 shows the aesthetic classification accuracy of the individual dgCNN models (Europe, USA, Female, Male, 50+, 50−) on their respective subsets of the AVA-Portraits+ dataset. The red markers indicate the average accuracy (across 5 folds) while the blue lines indicate the dispersion range (minimum to maximum). For comparison, we also furnish the results of the same models trained with randomly sampled photos of the same attribute. For all demographic attributes, the the classification performance of models trained with demographic specific data performed slightly better than models trained with random photos. The "USA" and "Female" attributes appear to have strong characteristics that relied on demographic-oriented samples to improve against random training samples. Attributes with larger dispersion error such as "Europe", "Female" and "Age 50+" could indicate more ambiguity in terms of what constitutes a good portrait.

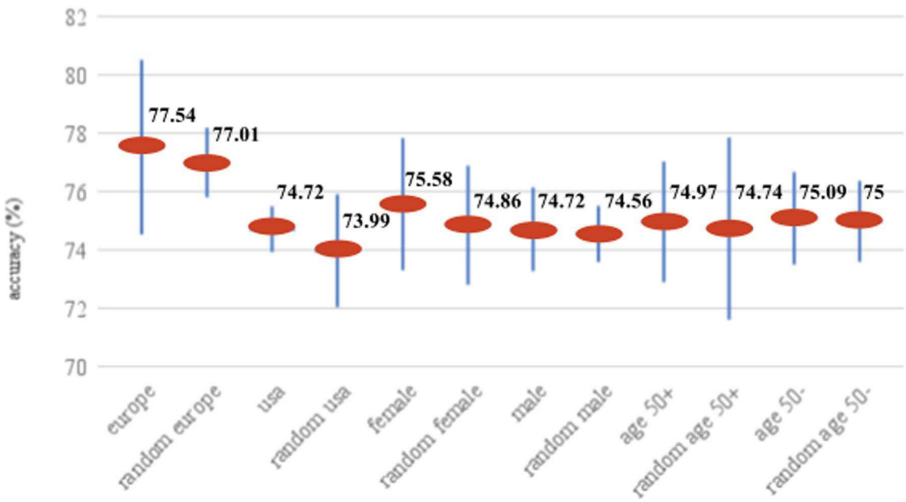

Fig. 6. Classification accuracy and dispersion errors of various dgCNN models and their respective models trained with random photos. (Color figure online)

Table 3 reports the state-of-the-art performance of our proposed MdgCNN architecture in comparison with other deep learning approaches in predicting aesthetics of portraitures. We achieve the best performance of 73.70% using the FC7 activations from the learned dgCNNs, outperforming our deep learning baseline by almost 5% albeit a marginally higher dispersion error. In both cases, adding a drop-out layer actually resulted in slightly poorer accuracy. It is crucial to mention that the results in Table 3 are for the evaluation of the entire AVA-Portraits+ dataset while results in Fig. 6 are only for each demographic subset, as outlined in Fig. 2.

Table 3. Aesthetic classification accuracy of the proposed MdgCNN method in comparison with other deep learning approaches.

Methods	Accuracy (%)	Error ± (%)
Kairanbay et al. [16]	65.33	1.02
Fine-tuned AlexNet (baseline)	68.46	1.95
MdgCNN (Merged FC6 w. drop-out)	73.10	2.92
MdgCNN (Merged FC6)	73.32	2.95
MdgCNN (Merged FC7 w. drop-out)	73.48	3.30
MdgCNN (Merged FC7)	**73.70**	3.08

Interestingly, the neurons from a specific dgCNN models are strongly activated if they contribute towards a portrait of high aesthetic score; this is akin to the MdgCNN responding towards a particular demographic attribute, and vice versa. Figure 7 depicts two sample photos of high and low aesthetic scores and their corresponding merged FC7 activations. The following figure (Fig. 8) shows some sample misclassified portrait photos. We observe that most of the false negatives (aesthetically high but misclassified as low) have unique composition or partial faces, while the false positives (aesthetically low but misclassified as high) used special effect filters such as sepia and grayscale.

Fig. 7. Merged FC7 activations of dgCNNs for two sample photos. Ticks denote the photographer's profile.

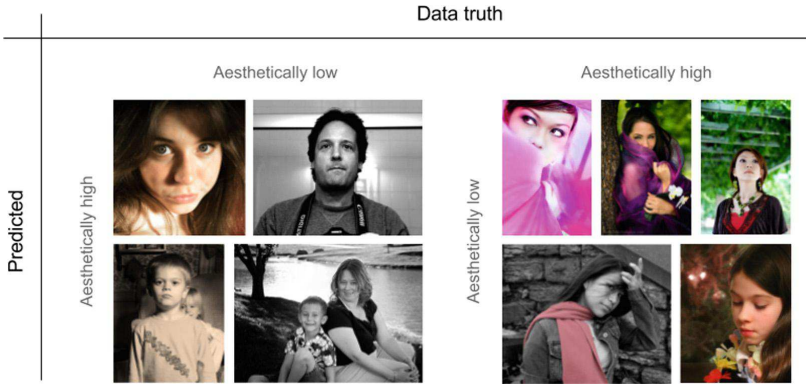

Fig. 8. Sample misclassified portraits.

Future Directions. This work presents an exploratory attempt at understanding the essence of individuality in the aesthetic quality of images, particularly portraitures, from the computational perspective. Current techniques have disregarded the aspect of the photographer which plays an important role, as we have shown in our experiments. We find Jacobsen's [1] observation of "differences in age and gender, within cultures" as a potential direction towards a more effective computational model. Our model lacks this hierarchical processing, as the simplistic merging of the dgCNNs can be improved. Also, unsupervised methods [20,21] that can capture latent information from these natural demographical tendencies, could be useful for learning disentangled factors.

Therefore, we would like to invite the research community to work on this aspect by making publicly available the AVA-Portraits+ demographic meta-data that we have extracted[2]. We also intend to release the similar meta-data for the full AVA dataset in the near future.

6 Conclusion

This paper investigates the influence of the photographer's demographic attributes such as location, age and gender on portrait photos. In order to do that, we extended the AVA-Portraits subset of the AVA dataset to include new demographical meta-data. We proposed to train demographic-based CNN models from sampled subsets of the data. By fusing together these individual models into a higher level network, we achieve state-of-the-art performance for aesthetic prediction of portraits, outperforming the vanilla baseline solution by around 5%. We envision great potential in the practical use of demographic-centric models to capture the natural inclinations of photographers of different profiles. In future, we intend to explore such multi-tiered architectures for other information such as image style, semantics and textual comments.

[2] AVA-Portraits+ demographic meta-data is available at https://goo.gl/jo13f3.

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